

Portfolio Construction and Factor Research

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An academic Python study of allocation, exploratory equity factors and Black-Litterman views. The public portfolio separates the original coursework from a newly tested publication revision. Licensed market data, assessment questions and private submission documents are excluded.

Original research scope

Component	Documented scope
Dataset	525 weekly price observations, October 2009-October 2019
Universe	10 US equities and three market benchmarks
Portfolio exploration	10,000 randomly generated long-only allocations
Factor diagnostics	HML/SMB-style spreads and 60-week rolling regressions
Black-Litterman	Equilibrium returns, investor views and three risk-aversion settings

Research contribution

The original notebooks load and transform weekly prices, compare return/covariance-based allocations and investigate exploratory factor exposures. The Black-Litterman exercise combines equilibrium assumptions with view matrices and uncertainty, translating different assumptions into allocation choices.

Evidence available for review

The archived notebooks preserve the original numerical source, with outputs and execution metadata removed. The public repository adds an input schema, a source-to-claim map and a separately dated revision with runnable tests. The original report identifies Wind as its market-data source; the licensed observations are not redistributed.

Interpretation

The original code contains limitations involving risk-free conventions, drawdown calculation, end-of-sample characteristics and incomplete factor-regression specifications. The archive is retained transparently. No original Sharpe, drawdown, trading alpha or live-return figure is presented as independently verified. The 2026 corrections below are new work, not claims about what was submitted in 2024.

Tested 2026 publication revision

AI-assisted implementation and verification, 12 September 2026

A new, self-contained Python module corrects or explicitly constrains the issues identified during review. It requires no data subscription or network access after installing the pinned dependencies. Original source files remain unchanged.

Control	Revised implementation
Return units	Separate effective annual-rate conversion and explicitly approximate simple-yield accrual
Information timing	Strict prior-availability joins, stale-data rejection and point-in-time characteristic selection
Risk measurement	Wealth-index drawdown including initial capital; aligned excess-return Sharpe
Model evaluation	Seeded long-only Monte Carlo candidates; full market/SMB/HML OLS; tested Black-Litterman posterior
Execution sequence	Completed-period training, next-period evaluation and drift-adjusted transaction-cost calculation

Verification performed

28 unit and regression tests passed locally. Checks include independently computed rate conversions and Bayesian updates, total-loss drawdown, factor-coefficient recovery, and future-data perturbation tests. A complete synthetic demonstration ran twice and returned identical result dictionaries.

Runnable demonstration

The fixture generates 10 artificial assets over 104 weekly return periods. The example uses 52 completed periods for training, then evaluates 52 periods with 10,000 candidate allocations per decision. Costs equal 10 basis points per unit of gross traded notional. These are software-test settings; generated performance figures are explicitly labeled synthetic and are not investment evidence.

Remaining limits

The module is an educational research framework, not production software. It does not reconstruct licensed historical inputs, instrument-level Treasury returns, a point-in-time investable universe, slippage, delistings or realistic execution delays. The example assumes prior-close execution. Original-market performance has not been independently reproduced.

See revised/README.md for commands and method references, revised/TEST_RESULTS.md for executed checks, and docs/EVIDENCE_AND_LIMITATIONS.md for archival source issues.